

Reproducing Dobi-SVD: A Reproduction and Evaluation Across Models

Optimization Class Project. MIPT

Kabirov Ruslan, Kabirov Rustam, Zahodov Egor, Ovseenko Ilya

Introduction

As large language models continue to grow, their compression has become an increasingly important problem in AI. SVD is a natural approach to model compression, but plain SVD often yields poor results. Recent activation-aware methods, such as ASVD, LLM-SVD, and ResSVD, substantially improve SVD-based compression. Nevertheless, these methods still suffer from significant performance degradation at high compression ratios. Dobi-SVD effectively addresses this issue and demonstrates strong benchmark performance. We reproduce Dobi-SVD and compare it against activation-aware methods on several models and tasks.

The idea behind activation-aware approaches

The goal is not to approximate the weight matrix itself, but to preserve the layer outputs on real calibration data:

$$\underbrace{\min_{W'} \|W - W'\|_F^2}_{\text{weight-aware objective}} \implies \underbrace{\min_{W'} \|XW - XW'\|_F^2}_{\text{activation-aware objective}},$$

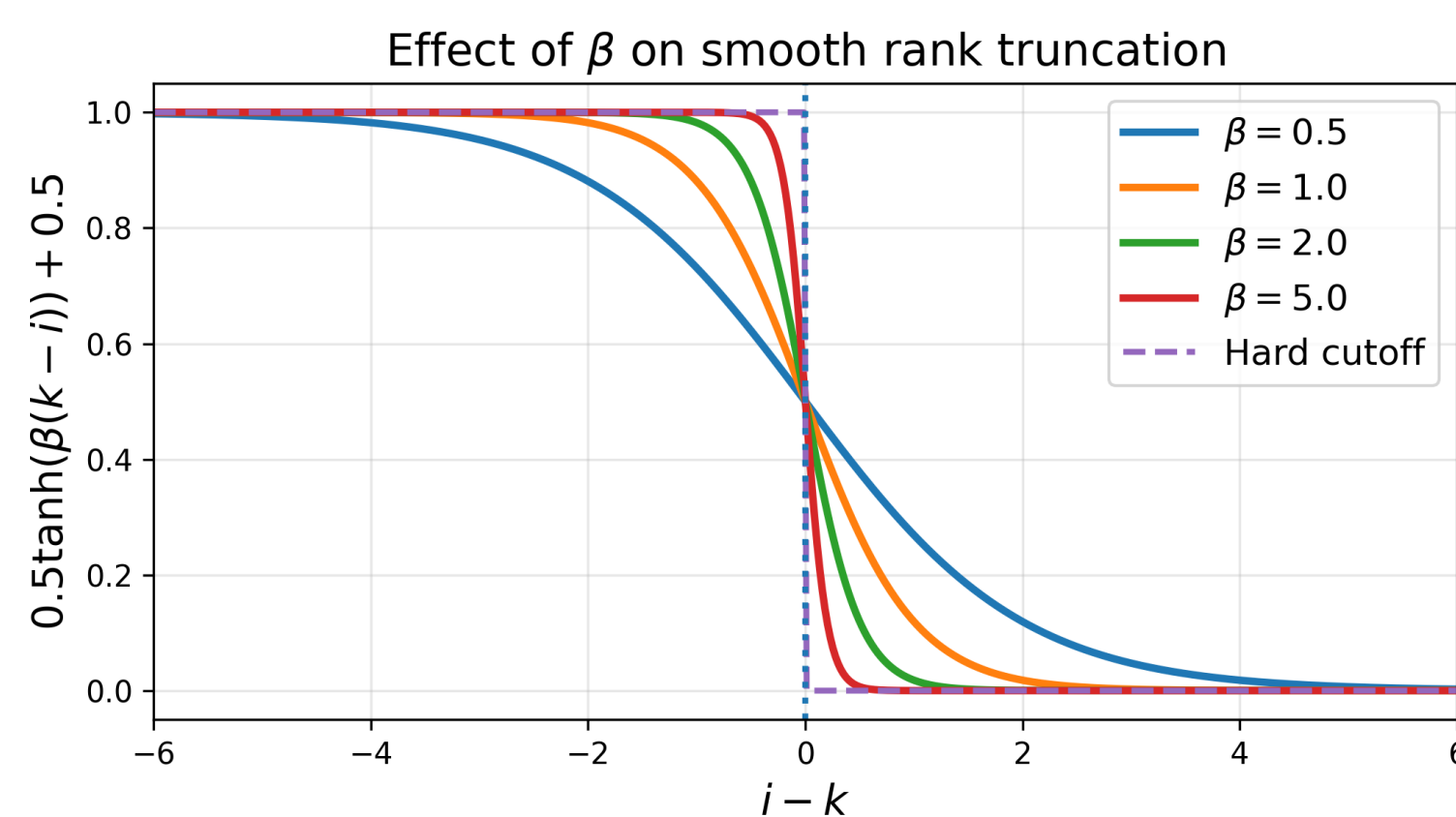
where X is the matrix of input activations. Although this problem admits a closed-form solution, it remains a local layer-wise objective. A **locally optimal approximation of each layer does not guarantee that the compressed model will preserve its end-to-end behavior**.

How does Dobi-SVD tackle this limitation?

They introduce a trainable effective rank using a smooth $\tanh$ -based threshold. Each singular value σ_i is transformed as

$$T(\sigma_i) = \sigma_i [0.5 \tanh(\beta(k - i)) + 0.5],$$

where the trainable parameter k controls how many singular values are retained. The $\tanh$ gate acts as a smooth approximation of hard rank truncation: singular values before the cutoff are mostly preserved, while those after the cutoff are suppressed.



This differentiable approximation turns rank selection into a numerical optimization problem. Dobi-SVD minimizes the objective

$$\mathcal{L} = \mathcal{L}_{\text{original}} + \lambda |R_{\text{now}}(k) - R_{\text{tar}}|,$$

where λ is a hyperparameter, $R_{\text{now}}(k)$ is the current compression ratio induced by the learnable ranks k , and R_{tar} is the target compression ratio.

The algorithm

Algorithm 1: Differentiable search for optimal k

Input: Training data $\{x_1, \dots, x_n\}$, target compression ratio R_{tar} .

Output: Optimal k values for each weight matrix W .

Step 1: Smooth activation truncation inference.

1: **for each** activation A **do**

2: Apply SVD:

$$A = U_A \Sigma_A V_A^\top, \quad \Sigma_A = [\sigma_1, \dots, \sigma_n].$$

3: Get smooth truncation:

$$T(\sigma_i) = \sigma_i \left[\frac{1}{2} \tanh(\beta(k - i)) + \frac{1}{2} \right], \quad \Sigma_k = [T(\sigma_1), \dots, T(\sigma_n)].$$

4: Reconstruct activations:

$$\tilde{A} = U_A \Sigma_k V_A^\top.$$

5: **end for**

Step 2: Multi-objective loss training.

Freeze all parameters and keep only k trainable.

6: **for each** training step **do**

7: Compute current compression ratio R_{now} based on current k .

8: Compute multi-objective loss:

$$L = L_{\text{task}} + \lambda |R_{\text{now}} - R_{\text{tar}}|.$$

9: Update k values by minimizing L .

10: **end for**

Back propagation

We have $A = U \Sigma V^T$. Let $g_U = \partial L / \partial U$, $g_\Sigma = \partial L / \partial \Sigma$, and $g_V = \partial L / \partial V$. Our goal is to compute $\partial L / \partial A$.

Taking the differential of the SVD decomposition gives

$$dA = dU \Sigma V^T + U d\Sigma V^T + U \Sigma dV^T.$$

Multiplying by U^T from the left and by V from the right, we get

$$B := U^T dA V = U^T dU \Sigma + d\Sigma + \Sigma dV^T V.$$

Since U and V are orthogonal, $U^T dU$ and $dV^T V$ are skew-symmetric. Hence their diagonal entries are zero, so the diagonal part of B is simply $B_{ii} = d\sigma_i$.

For $i \neq j$, denote $a = (U^T dU)_{ij}$ and $b = (dV^T V)_{ij}$. Then

$$B_{ij} = a\sigma_j + \sigma_i b, \quad B_{ji} = -a\sigma_i - \sigma_j b.$$

Solving this 2×2 system introduces the denominator $\sigma_j^2 - \sigma_i^2$. Therefore, the SVD backward pass contains terms of the form

$$\frac{1}{\sigma_j^2 - \sigma_i^2}.$$

When two singular values are close, this denominator becomes very small, which may lead to exploding gradients. Dobi-SVD stabilizes this step by replacing the unstable term with a bounded Taylor-based approximation.

Experiments

This work focuses on benchmarking Dobi-SVD against existing SVD-based compression methods for memory-efficient LLM inference. We compare Dobi-SVD with LLM-SVD, RES-SVD, and ASVD on LLaMA-7B, quantized LLaMA-7B, and LLaMA-1B. Experiments are conducted on WikiText-2, PTB, and C4, with perplexity as the primary metric. The goal is to evaluate whether Dobi-SVD provides a better trade-off between compression and language modeling quality across model scales and quantization settings.

Results

Ratio	Method	LLaMA-7B		LLaMA-1B		LLaMA-7B + GPTQ	
		Wiki2	PTB C4	Wiki2	PTB C4	Wiki2	PTB C4
1.0	Baseline	5.46	— —	— — —	— — —	— — —	—
0.9	ASVD	5.97	— —	— — —	— — —	— — —	—
	RES-SVD	6.04	— —	— — —	— — —	— — —	—
	SVD-LLM	7.29	— —	— — —	— — —	— — —	—
	Dobi-SVD	7.26	— —	— — —	— — —	— — —	—
0.8	ASVD	9.74	— —	— — —	— — —	— — —	—
	RES-SVD	8.00	— —	— — —	— — —	— — —	—
	SVD-LLM	8.33	— —	— — —	— — —	— — —	—
	Dobi-SVD	8.37	— —	— — —	— — —	— — —	—
0.7	ASVD	1245	— —	— — —	— — —	— — —	—
	RES-SVD	10.09	— —	— — —	— — —	— — —	—
	SVD-LLM	10.53	— —	— — —	— — —	— — —	—
	Dobi-SVD	10.44	— —	— — —	— — —	— — —	—

Table 1: Perplexity comparison of SVD-based compression methods across model scales and quantization settings.

Conclusion

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Acknowledgements

This material is based upon work supported by the X Fellowship and my mom.